

Statistiques Descriptives - StatApp

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Importation des librairies et des bases de données

On commence par importer les librairies et les bases de données nécessaires.

```
library(haven)
library(psych)
library(ggplot2)
library(patchwork)
library(tidyr)
library(dplyr)
library(sandwich)
library(lmtest)

df <- read_dta("waveI_young_women_sample.dta")
df_2 <- read_dta("waveIII_young_women_sample.dta")
df_all <- read_dta("waveIII.dta")
df_hh <- read_dta("waveI_households.dta")
```

Première vague : wave I (baseline)

Aperçu général

```
# head(df)
dim(df)      # 11350 obs, 26 variables
```

```
## [1] 11350    26
```

```
names(df)
```

```
## [1] "girlID" "bl_age_reported"
## [3] "bl_marriage_time_comfort" "bl_marriage_time_who"
## [5] "bl_earliest_marry" "bl_earliest_marry_why_phys"
## [7] "bl_latest_marry_why_dowry" "bl_latest_marry_why_groom"
## [9] "bl_latest_marry_why_gwill" "bl_latest_marry_why_parent"
## [11] "bl_latest_marry_why_child" "bl_latest_marry_why_earn"
## [13] "bl_latest_marry_why_other" "bl_latest_marry_why_dk"
## [15] "bl_desired_marriage_age" "bl_earlypreg"
## [17] "bl_risks_earlypreg_pain" "bl_risks_earlypreg_momrisk"
## [19] "bl_risks_earlypreg_childrisk" "bl_risks_earlypreg_momhealth"
## [21] "bl_risks_earlypreg_childhealth" "bl_risks_earlypreg_other"
## [23] "bl_ever_married" "bl_treatment_type"
```

```
## [25] "CLUSTER"                                "upcode"
```

Commentaires : La base de données initiale contient 11350 observations, soit 11350 filles interrogées lors de la première vague de questionnaire.

Dans le cadre de notre projet, il convient de nous intéresser à quelques variables en particulier : âge de mariage souhaité, éducation à la baseline, éducation de la mère... Ces données sont trouvables dans deux bases de données : la wave I et la wave III.

Stats descriptives globales

```
describe(df)
```

```
##               vars      n    mean      sd median trimmed
## girlID*         1 11350 5675.50 3276.61 5675.5 5675.50
## bl_age_reported  2 11350   13.16    2.15   13.0   13.11
## bl_marriage_time_comfort  3 11350    0.36    0.48    0.0    0.32
## bl_marriage_time_who    4  4046    4.35    2.24    3.0    4.27
## bl_earliest_marry    5 11350   21.19   14.60   18.0   18.42
## bl_earliest_marry_why_phys  6 10948    0.62    0.49    1.0    0.65
## bl_latest_marry_why_dowry  7 10622    0.25    0.43    0.0    0.19
## bl_latest_marry_why_groom  8 10622    0.27    0.44    0.0    0.21
## bl_latest_marry_why_gwill  9 10622    0.38    0.48    0.0    0.34
## bl_latest_marry_why_parent 10 10622    0.13    0.34    0.0    0.04
## bl_latest_marry_why_child 11 10622    0.04    0.19    0.0    0.00
## bl_latest_marry_why_earn 12 10622    0.01    0.09    0.0    0.00
## bl_latest_marry_why_other 13 10622    0.00    0.03    0.0    0.00
## bl_latest_marry_why_dk    14 10622    0.32    0.46    0.0    0.27
## bl_desired_marriage_age   15 10268   20.44    2.72   20.0   20.15
## bl_earlypreg          16 11348    0.55    0.50    1.0    0.57
## bl_risks_earlypreg_pain   17 11349    0.05    0.21    0.0    0.00
## bl_risks_earlypreg_momrisk 18 11349    0.38    0.48    0.0    0.34
## bl_risks_earlypreg_childrisk 19 11349    0.26    0.44    0.0    0.20
## bl_risks_earlypreg_momhealth 20 11349    0.29    0.45    0.0    0.23
## bl_risks_earlypreg_childhealth 21 11349    0.16    0.37    0.0    0.08
## bl_risks_earlypreg_other  22 11349    0.00    0.04    0.0    0.00
## bl_ever_married         23 11350    0.02    0.15    0.0    0.00
## bl_treatment_type       24 11350    2.50    1.25    3.0    2.50
## CLUSTER                25 11350   232.64   133.19   234.0   233.16
## upcode                 26 11350   47.48   33.73   38.0   47.10
##
##               mad min    max range  skew kurtosis    se
## girlID*        4206.88    1 11350 11349    0.00   -1.20 30.76
## bl_age_reported    2.97   10    17     7    0.16   -1.10  0.02
## bl_marriage_time_comfort  0.00    0     1     1    0.60   -1.64  0.00
## bl_marriage_time_who    1.48    1    11    10    0.38   -1.29  0.04
## bl_earliest_marry    0.00    9    97    88    4.94   22.68  0.14
## bl_earliest_marry_why_phys  0.00    0     1     1   -0.49   -1.76  0.00
## bl_latest_marry_why_dowry  0.00    0     1     1    1.14   -0.70  0.00
## bl_latest_marry_why_groom  0.00    0     1     1    1.06   -0.88  0.00
## bl_latest_marry_why_gwill  0.00    0     1     1    0.51   -1.74  0.00
## bl_latest_marry_why_parent  0.00    0     1     1    2.17    2.71  0.00
## bl_latest_marry_why_child  0.00    0     1     1    4.82   21.26  0.00
## bl_latest_marry_why_earn    0.00    0     1     1   10.49   107.99  0.00
```

```
## bl_latest_marry_why_other      0.00  0    1    1 36.39 1322.50  0.00
## bl_latest_marry_why_dk         0.00  0    1    1  0.79   -1.37  0.00
## bl_desired_marriage_age        2.97 11   40   29  1.20    2.52  0.03
## bl_earlypreg                   0.00  0    1    1 -0.21   -1.95  0.00
## bl_risks_earlypreg_pain        0.00  0    1    1  4.26   16.14  0.00
## bl_risks_earlypreg_momrisk     0.00  0    1    1  0.52   -1.73  0.00
## bl_risks_earlypreg_childrisk   0.00  0    1    1  1.07   -0.84  0.00
## bl_risks_earlypreg_momhealth   0.00  0    1    1  0.94   -1.11  0.00
## bl_risks_earlypreg_childhealth 0.00  0    1    1  1.81    1.27  0.00
## bl_risks_earlypreg_other       0.00  0    1    1 24.38   592.21  0.00
## bl_ever_married                0.00  0    1    1  6.33   38.01  0.00
## bl_treatment_type              1.48  1    4    3  0.00   -1.63  0.01
## CLUSTER                       173.46  1  460  459 -0.02   -1.22  1.25
## upcode                        45.96  3   95   92  0.20   -1.42  0.32
```

```
# Valeurs manquantes par variable
colSums(is.na(df))
```

```
##                girlID                bl_age_reported
##                0                0
##    bl_marriage_time_comfort    bl_marriage_time_who
##                0                7304
##    bl_earliest_marry    bl_earliest_marry_why_phys
##                0                402
##    bl_latest_marry_why_dowry    bl_latest_marry_why_groom
##                728                728
##    bl_latest_marry_why_gwill    bl_latest_marry_why_parent
##                728                728
##    bl_latest_marry_why_child    bl_latest_marry_why_earn
##                728                728
##    bl_latest_marry_why_other    bl_latest_marry_why_dk
##                728                728
##    bl_desired_marriage_age                bl_earlypreg
##                1082                2
##    bl_risks_earlypreg_pain    bl_risks_earlypreg_momrisk
##                1                1
##    bl_risks_earlypreg_childrisk    bl_risks_earlypreg_momhealth
##                1                1
##    bl_risks_earlypreg_childhealth    bl_risks_earlypreg_other
##                1                1
##                bl_ever_married                bl_treatment_type
##                0                0
##                CLUSTER                upcode
##                0                0
```

Peu de valeurs manquantes pour les variables qui nous intéressent.

Statistiques sur l'âge

```
mean(df$bl_age_reported, na.rm = TRUE)      # ~13.2
```

```
## [1] 13.15894
```

```
sd(df$bl_age_reported, na.rm = TRUE)

## [1] 2.147714
median(df$bl_age_reported, na.rm = TRUE)

## [1] 13
range(df$bl_age_reported, na.rm = TRUE)      # 10 - 17

## [1] 10 17
table(df$bl_age_reported)                    # distribution par âge

##
##    10    11    12    13    14    15    16    17
## 1499 1590 1715 1638 1477 1366 1273  792
```

On remarque que les filles interrogées ont entre 10 et 17 ans, et ont en moyenne 13 ans.

Âge souhaité au mariage

```
mean(df$bl_desired_marriage_age, na.rm = TRUE)

## [1] 20.444
sd(df$bl_desired_marriage_age, na.rm = TRUE)

## [1] 2.71845
table(df$bl_desired_marriage_age)

##
##    11    12    13    14    15    16    17    18    19    20    21    22    23    24    25    26
##     2    12     6    18   112   107   122 2401   317 4354   316   751   137    95 1222    73
##    27    28    29    30    32    35    40
##    14    51     4   145     1     5     3
```

En moyenne, les filles souhaitent se marier à 20 ans, avec une forte proportion à 25 ans également. Cela ne reflète pas la réalité du mariage infantile, montrant que les jeunes filles ont sûrement peu de choix dans leur avenir matrimonial.

Variables binaires / catégorielles

```
# Déjà mariée
table(df$bl_ever_married)

##
##      0      1
## 11086   264

prop.table(table(df$bl_ever_married)) * 100      # en %

##
##      0      1
## 97.674009  2.325991
```

```
# Type de traitement
table(df$bl_treatment_type)
```

```
##
##      1      2      3      4
## 3731 1938 1958 3723
```

```
prop.table(table(df$bl_treatment_type)) * 100
```

```
##
##      1      2      3      4
## 32.87225 17.07489 17.25110 32.80176
```

264 filles sont déjà mariées (tableau croisé avec l'âge de ces filles dans la partie 9). On voit que l'assignation au traitement suit bien le ratio présenté dans l'article.

Stats par groupe

```
# Âge moyen selon le type de traitement
aggregate(bl_age_reported ~ bl_treatment_type, data = df, FUN = mean)
```

```
##   bl_treatment_type bl_age_reported
## 1                   1         13.15117
## 2                   2         13.12436
## 3                   3         13.21859
## 4                   4         13.15337
```

```
# Âge moyen selon déjà mariée ou non
aggregate(bl_age_reported ~ bl_ever_married, data = df, FUN = mean)
```

```
##   bl_ever_married bl_age_reported
## 1                0         13.09544
## 2                1         15.82576
```

C'est ce qu'on retrouve ensuite : les filles déjà mariées ont pour la grande majorité 17 ans (et sont donc à un âge plus avancé).

```
# Âge souhaité au mariage selon le traitement
aggregate(bl_desired_marriage_age ~ bl_treatment_type,
          data = df, FUN = mean, na.rm = TRUE)
```

```
##   bl_treatment_type bl_desired_marriage_age
## 1                   1         20.42383
## 2                   2         20.48540
## 3                   3         20.40353
## 4                   4         20.46372
```

```
# Stats complètes par groupe
aggregate(bl_age_reported ~ bl_treatment_type, data = df,
          FUN = function(x) c(mean = mean(x), sd = sd(x), n = length(x)))
```

```
##   bl_treatment_type bl_age_reported.mean bl_age_reported.sd bl_age_reported.n
## 1                   1         13.151166         2.123451      3731.000000
## 2                   2         13.124355         2.148468      1938.000000
## 3                   3         13.218590         2.169426      1958.000000
## 4                   4         13.153371         2.160253      3723.000000
```

On remarque que les caractéristiques par traitement sont bien équilibrées par rapport à l'âge à la baseline et l'âge de mariage souhaité.

Âge souhaité au mariage $\times$ âge des filles

```
# Âge souhaité au mariage selon l'âge des filles
aggregate(bl_desired_marriage_age ~ bl_age_reported,
          data = df, FUN = mean, na.rm = TRUE)
```

```
##   bl_age_reported bl_desired_marriage_age
## 1                10                20.02304
## 2                11                20.13582
## 3                12                20.17859
## 4                13                20.48440
## 5                14                20.58976
## 6                15                20.70223
## 7                16                20.82929
## 8                17                21.02083
```

```
# Stats complètes par âge
aggregate(bl_desired_marriage_age ~ bl_age_reported, data = df,
          FUN = function(x) c(mean = mean(x, na.rm = TRUE),
                               sd   = sd(x, na.rm = TRUE),
                               n     = length(x)))
```

```
##   bl_age_reported bl_desired_marriage_age.mean bl_desired_marriage_age.sd
## 1                10                20.023041                2.816623
## 2                11                20.135820                2.678394
## 3                12                20.178594                2.721044
## 4                13                20.484395                2.694862
## 5                14                20.589762                2.700486
## 6                15                20.702229                2.711732
## 7                16                20.829290                2.657886
## 8                17                21.020833                2.544890
##   bl_desired_marriage_age.n
## 1                1302.000000
## 2                1421.000000
## 3                1579.000000
## 4                1538.000000
## 5                1387.000000
## 6                1256.000000
## 7                1113.000000
## 8                672.000000
```

```
# Tableau croisé : tranches d'âge  $\times$  âge souhaité au mariage
df$tranche_age <- cut(df$bl_age_reported,
                      breaks = c(9, 11, 13, 15, 17),
                      labels = c("10-11", "12-13", "14-15", "16-17"))
```

```
aggregate(bl_desired_marriage_age ~ tranche_age,
          data = df, FUN = mean, na.rm = TRUE)
```

```
##   tranche_age bl_desired_marriage_age
## 1      10-11                20.08189
```

```
## 2      12-13      20.32948
## 3      14-15      20.64321
## 4      16-17      20.90140
```

Commentaire : Finalement, il n'y a pas vraiment d'impact sur l'âge du mariage souhaité et la tranche d'âge de la personne interrogée. L'âge de mariage souhaité augmente au fil de l'âge.

Visualisation de l'âge de mariage vs le taux de filles mariées à la baseline

```
# Tableau croisé : tranches d'âge x âge souhaité au mariage
df$tranche_age <- cut(df$bl_age_reported,
breaks = c(9, 11, 13, 15, 17),
labels = c("10-11", "12-13", "14-15", "16-17"))
aggregate(bl_desired_marriage_age ~ tranche_age,
data = df, FUN = mean, na.rm = TRUE)

##   tranche_age bl_desired_marriage_age
## 1      10-11      20.08189
## 2      12-13      20.32948
## 3      14-15      20.64321
## 4      16-17      20.90140

prop_mariees <- tapply(df$bl_ever_married, df$tranche_age,
mean, na.rm = TRUE) * 100

# Graphiques côte à côte

# *** PRÉPARATION DES DONNÉES ***
df_prop <- data.frame(
  tranche = names(prop_mariees),
  pourcentage = as.numeric(prop_mariees)
)

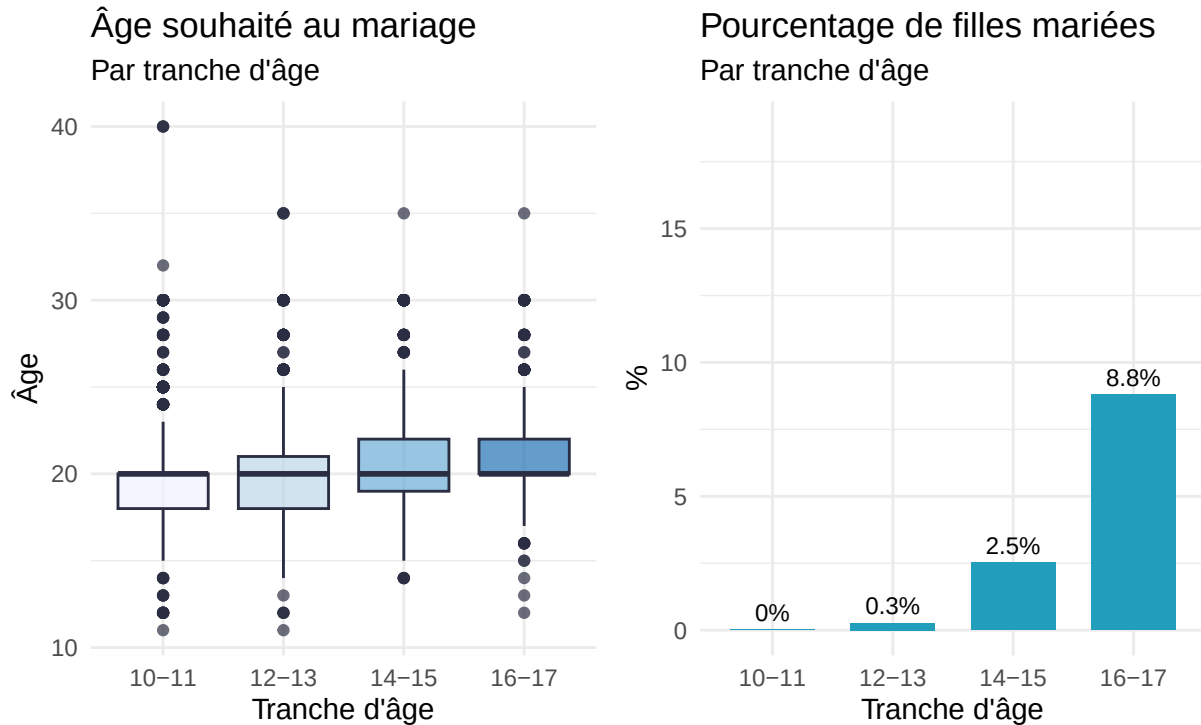
# *** GRAPHIQUE 1 : Boxplot ***
p1 <- ggplot(df, aes(x = tranche_age, y = bl_desired_marriage_age, fill = tranche_age)) +
  geom_boxplot(alpha = 0.7, color = "#2b2d42") +
  scale_fill_brewer(palette = "Blues") + # Palette automatique de bleus
  theme_minimal() +
  labs(title = "Âge souhaité au mariage",
        subtitle = "Par tranche d'âge",
        x = "Tranche d'âge", y = "Âge") +
  theme(legend.position = "none")

# *** GRAPHIQUE 2 : Barplot ***
p2 <- ggplot(df_prop, aes(x = tranche, y = pourcentage)) +
  geom_bar(stat = "identity", fill = "#219ebc", width = 0.7) +
  geom_text(aes(label = paste0(round(pourcentage, 1), "%")), vjust = -0.5, size = 3) +
  theme_minimal() +
  labs(title = "Pourcentage de filles mariées",
        subtitle = "Par tranche d'âge",
        x = "Tranche d'âge", y = "%") +
  scale_y_continuous(limits = c(0, max(df_prop$pourcentage) + 10))
```

```
# *** ASSEMBLAGE CÔTE À CÔTE ***
p1 + p2 + plot_annotation(title = "Analyse comparative du mariage par tranche d'âge",
  caption = "Note : Le graphique a été réalisé à partir de l'échantillon large",
  theme = theme(plot.title = element_text(size = 16, face = "bold", hjust = 0.5))

## Warning: Removed 1082 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

Analyse comparative du mariage par tranche d'âge



Le graphique a été réalisé à partir de l'échantillon large comprenant 11 350 jeunes filles (âgées de 10 à 17 ans).

Constitution de df_end

On s'intéresse à la dernière vague : dataframe plus complète en terme de variables. On crée donc df_end afin de mener des statistiques sur l'échantillon young women's sample. Conformément au choix des auteurs, on garde seulement les filles qui avaient entre 15 et 17 ans et qui n'étaient pas mariées à la baseline.

Import & double jointure

```
# Charger les trois fichiers
df      <- read_dta("waveI_young_women_sample.dta")
df_end  <- read_dta("waveIII_young_women_sample.dta")
df_wave3 <- df_all

# Variables à récupérer depuis wave_III
vars_w3 <- c("girlID",
```



```

    "matchmaker_3", "has_primary", "has_secondary",
    "bl_education", "bl_rural", "bl_girl_boy_ratio",
    "bl_public_transit",
    "hh_village",
    "secondary_complete", "age_gap",
    "arranged_marriage",
    "outside_village",
    "husband_formal", "outside_union",
    "distance_vc_ss",
    "matchmaker", "el_age_predicted")

# Variables à récupérer depuis Wave I
vars_w1 <- c("girlID",
            "bl_desired_marriage_age", # + ajouté ici (vient du Wave I !)
            "bl_earlypreg",
            "bl_treatment_type")

# Jointure 1 : df_end + Wave I (baseline)
df_end <- merge(df_end, df[, vars_w1], by = "girlID", all.x = TRUE)

# Jointure 2 : df_end + wave_III (variables manquantes)
df_end <- merge(df_end, df_wave3[, vars_w3], by = "girlID", all.x = TRUE)

# Résoudre les doublons .x / .y après merge
doublons <- c("arranged_marriage", "bl_public_transit", "distance_vc_ss",
            "bl_desired_marriage_age")

for (var in doublons) {
  if (paste0(var, ".x") %in% names(df_end)) {
    df_end[[var]] <- df_end[[paste0(var, ".x")]]
    df_end[[paste0(var, ".x")]] <- NULL
    df_end[[paste0(var, ".y")]] <- NULL
  }
}

# Filtre : 15-17 ans au départ + non mariées à la baseline + suivies à l'endline
df_end <- df_end[df_end$bl_age_reported >= 15 & df_end$bl_age_reported <= 17 &
                df_end$bl_ever_married == 0 &
                df_end$endline == 1, ]

# Conversion des colonnes haven_labelled en numeric/factor
df_end <- as.data.frame(lapply(df_end, function(x) {
  if (inherits(x, "haven_labelled")) as.numeric(x) else x
})))

# Renommer si merge a créé des doublons
if ("bl_desired_marriage_age.x" %in% names(df_end)) {
  df_end$bl_desired_marriage_age <- df_end$bl_desired_marriage_age.x
  df_end$bl_desired_marriage_age.x <- NULL
  df_end$bl_desired_marriage_age.y <- NULL
}

```

```
}

dim(df_end)    # ~1821 obs
```

```
## [1] 1821 167
```

L'échantillon final est composé des 1821 filles (entre 15 et 17 ans, non mariées au début de l'expérience).

Sélection des variables finales

```
vars_finales <- c("girlID", "bl_age_reported", "treatment_type",
  # Éducation
  "bl_education", "education",
  "bl_still_in_school", "still_in_school",
  "secondary_complete",
  # Ménage & famille
  "bl_HHsize", "older_sister",
  "bl_mother_schooled", "bl_girl_boy_ratio",
  # Mariage
  "ever_married", "marriage_age",
  "under_18", "under_16",
  "age_married_desired", "bl_desired_marriage_age",
  "arranged_marriage", "age_gap",
  # Infrastructure village
  "has_primary", "has_secondary",
  "matchmaker_3", "bl_rural",
  "bl_public_transit", "distance_vc_ss",
  # Autres variables
  "outside_village", "outside_union",
  "husband_formal", "hh_village", "high_g_cons",
  "high_p_cons",
  "marriage_age", "ever_birth_20", "ever_worked", "bl_income", "CLUSTER")

df_end <- df_end[, vars_finales]
summary(df_end)
```

```
##      girlID      bl_age_reported treatment_type  bl_education
## Length:1821      Min.      :15.00      Min.      :1.000      Min.      : 0.000
## Class :character  1st Qu.:15.00      1st Qu.:1.000      1st Qu.: 5.000
## Mode  :character  Median :16.00      Median :3.000      Median : 8.000
##                               Mean  :15.71      Mean  :2.532      Mean  : 7.052
##                               3rd Qu.:16.00      3rd Qu.:4.000      3rd Qu.: 9.000
##                               Max.   :17.00      Max.   :4.000      Max.   :11.000
##                               NA's    :342
##      education      bl_still_in_school still_in_school  secondary_complete
## Min.      : 0.000      Min.      :0.0000      Min.      :0.0000      Min.      :0.0000
## 1st Qu.: 6.000      1st Qu.:0.0000      1st Qu.:0.0000      1st Qu.:0.0000
## Median :10.000      Median :1.0000      Median :0.0000      Median :0.0000
## Mean  : 9.711      Mean  :0.6376      Mean  :0.1741      Mean  :0.3556
## 3rd Qu.:14.000      3rd Qu.:1.0000      3rd Qu.:0.0000      3rd Qu.:1.0000
## Max.   :19.000      Max.   :1.0000      Max.   :1.0000      Max.   :1.0000
## NA's    :12              NA's    :12              NA's    :488
##      bl_HHsize      older_sister      bl_mother_schooled bl_girl_boy_ratio
```

```

## Min. : 0.000 Min. :0.0000 Min. :0.000 Min. :0.6923
## 1st Qu.: 3.000 1st Qu.:0.0000 1st Qu.:0.000 1st Qu.:1.3654
## Median : 5.000 Median :0.0000 Median :1.000 Median :1.7778
## Mean : 4.602 Mean :0.1225 Mean :0.635 Mean :2.0301
## 3rd Qu.: 6.000 3rd Qu.:0.0000 3rd Qu.:1.000 3rd Qu.:2.4167
## Max. :20.000 Max. :1.0000 Max. :1.000 Max. :9.5000
## NA's :884 NA's :339
## ever_married marriage_age under_18 under_16
## Min. :0.0000 Min. : 9.00 Min. :0.0000 Min. : NA
## 1st Qu.:1.0000 1st Qu.:16.00 1st Qu.:0.0000 1st Qu.: NA
## Median :1.0000 Median :17.00 Median :1.0000 Median : NA
## Mean :0.8988 Mean :17.32 Mean :0.5336 Mean :NaN
## 3rd Qu.:1.0000 3rd Qu.:19.00 3rd Qu.:1.0000 3rd Qu.: NA
## Max. :1.0000 Max. :25.81 Max. :1.0000 Max. : NA
## NA's :12 NA's :208 NA's :350 NA's :1821
## age_married_desired bl_desired_marriage_age arranged_marriage
## Min. : 8.00 Min. : 0.00 Min. :0.0000
## 1st Qu.:18.00 1st Qu.:18.00 1st Qu.:1.0000
## Median :20.00 Median :20.00 Median :1.0000
## Mean :20.29 Mean :19.42 Mean :0.9089
## 3rd Qu.:21.50 3rd Qu.:22.00 3rd Qu.:1.0000
## Max. :30.00 Max. :35.00 Max. :1.0000
## NA's :58 NA's :669
## age_gap has_primary has_secondary matchmaker_3
## Min. : -22.512 Min. :0.0000 Min. :0.0000 Min. :0.0000
## 1st Qu.: 2.406 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:0.0000
## Median : 5.513 Median :1.0000 Median :0.0000 Median :1.0000
## Mean : 5.379 Mean :0.5196 Mean :0.2679 Mean :0.7085
## 3rd Qu.: 6.661 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.:1.0000
## Max. :25.485 Max. :1.0000 Max. :1.0000 Max. :1.0000
## NA's :730 NA's :339 NA's :339 NA's :339
## bl_rural bl_public_transit distance_vc_ss outside_village
## Min. :0.00000 Min. :0.000 Min. : 6.0 Min. :0.0000
## 1st Qu.:0.00000 1st Qu.:0.000 1st Qu.:144.0 1st Qu.:1.0000
## Median :0.00000 Median :0.000 Median :366.0 Median :1.0000
## Mean :0.09042 Mean :0.296 Mean :677.3 Mean :0.9061
## 3rd Qu.:0.00000 3rd Qu.:1.000 3rd Qu.:823.0 3rd Qu.:1.0000
## Max. :1.00000 Max. :1.000 Max. :9748.0 Max. :1.0000
## NA's :339 NA's :671
## outside_union husband_formal hh_village high_g_cons
## Min. :0.0000 Min. :0.0000 Min. : 63.0 Min. :0.0000
## 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:193.8 1st Qu.:0.0000
## Median :0.0000 Median :0.0000 Median :311.0 Median :0.0000
## Mean :0.3557 Mean :0.4996 Mean :368.7 Mean :0.4195
## 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.:521.2 3rd Qu.:1.0000
## Max. :1.0000 Max. :1.0000 Max. :914.0 Max. :1.0000
## NA's :671 NA's :684 NA's :339
## high_p_cons marriage_age.1 ever_birth_20 ever_worked
## Min. :0.0000 Min. : 9.00 Min. :0.0000 Min. :0.0000
## 1st Qu.:0.0000 1st Qu.:16.00 1st Qu.:0.0000 1st Qu.:0.0000
## Median :0.0000 Median :17.00 Median :0.0000 Median :1.0000
## Mean :0.4315 Mean :17.32 Mean :0.2768 Mean :0.5601
## 3rd Qu.:1.0000 3rd Qu.:19.00 3rd Qu.:1.0000 3rd Qu.:1.0000
## Max. :1.0000 Max. :25.81 Max. :1.0000 Max. :1.0000

```

```
## NA's :32      NA's :208      NA's :11      NA's :16
## bl_income      CLUSTER
## Min. : 0.00    Min. : 1.0
## 1st Qu.: 10.25  1st Qu.:170.0
## Median : 16.75  Median :258.0
## Mean : 19.63    Mean :244.4
## 3rd Qu.: 23.93  3rd Qu.:356.0
## Max. :544.34    Max. :460.0
##
```

Commentaire : On ne garde que les variables qui seront intéressantes pour les tests d'hétérogénéité et sur lesquels la revue de la littérature nous permet de nous concentrer : l'éducation, la taille du ménage, les variables liées au mariage, l'infrastructure du village et d'autres variables.

Stats descriptives — variables continues

```
# Éducation baseline vs endline
data.frame(
  Moment = c("Baseline", "Endline"),
  Moyenne = c(mean(df_end$bl_education, na.rm = TRUE),
               mean(df_end$education, na.rm = TRUE)),
  Ecart_type = c(sd(df_end$bl_education, na.rm = TRUE),
                 sd(df_end$education, na.rm = TRUE))
)

##      Moment  Moyenne Ecart_type
## 1 Baseline  7.052062  2.501013
## 2 Endline   9.711443  4.656412

# Âge souhaité au mariage baseline vs endline
data.frame(
  Moment = c("Baseline", "Endline"),
  Moyenne = c(mean(df_end$bl_desired_marriage_age, na.rm = TRUE),
               mean(df_end$age_married_desired, na.rm = TRUE)),
  Ecart_type = c(sd(df_end$bl_desired_marriage_age, na.rm = TRUE),
                 sd(df_end$age_married_desired, na.rm = TRUE))
)

##      Moment  Moyenne Ecart_type
## 1 Baseline 19.42449  5.816463
## 2 Endline 20.29009  2.539924
```

Commentaire : Il est normal que l'âge soit de deux années plus élevées dû à l'étirement temporel de l'expérience. Concernant l'âge de mariage souhaité, il y a un léger décalage vers la droite.

```
# Taille du ménage
mean(df_end$bl_HHsize, na.rm = TRUE)

## [1] 4.602416

sd(df_end$bl_HHsize, na.rm = TRUE)

## [1] 2.817152

# Écart d'âge avec le mari (parmi mariées)
mean(df_end$age_gap, na.rm = TRUE)
```

```
## [1] 5.379111
```

Commentaire : Ecart d'âge d'environ 5 ans avec le mari et un ménage à la baseline constitué de 4 à 5 personnes.

Stats descriptives — variables binaires

```
vars_bin <- c("bl_still_in_school", "still_in_school", "secondary_complete",
             "older_sister", "bl_mother_schooled",
             "ever_married", "under_18", "under_16",
             "arranged_marriage", "matchmaker_3",
             "has_primary", "has_secondary",
             "bl_rural", "bl_public_transit",
             "outside_village", "husband_formal")

recap <- data.frame(
  Variable = vars_bin,
  Pct_oui  = sapply(df_end[, vars_bin],
                    function(x) round(mean(x, na.rm = TRUE) * 100, 1))
)
print(recap)
```

##		Variable	Pct_oui
##	bl_still_in_school	bl_still_in_school	63.8
##	still_in_school	still_in_school	17.4
##	secondary_complete	secondary_complete	35.6
##	older_sister	older_sister	12.2
##	bl_mother_schooled	bl_mother_schooled	63.5
##	ever_married	ever_married	89.9
##	under_18	under_18	53.4
##	under_16	under_16	NaN
##	arranged_marriage	arranged_marriage	90.9
##	matchmaker_3	matchmaker_3	70.9
##	has_primary	has_primary	52.0
##	has_secondary	has_secondary	26.8
##	bl_rural	bl_rural	9.0
##	bl_public_transit	bl_public_transit	29.6
##	outside_village	outside_village	90.6
##	husband_formal	husband_formal	50.0

Commentaire : 63.8% des filles interrogées sont toujours scolarisées au début de l'expérience et 63.5% ont une mère ayant poursuivi une éducation. A la fin de l'expérience, seules 17,4% des filles sont encore scolarisées et 89.9% d'entre elles sont mariées. Dans 90.9% des cas, le mariage a été arrangé (reregarder à quoi correspondent chaque variable).

Tableau de balance de la baseline

```
mean(df_end$bl_age_reported, na.rm = TRUE)
```

```
## [1] 15.7084
```

```

mean(df_end$bl_still_in_school, na.rm = TRUE)

## [1] 0.6375618
mean(df_end$bl_HHsize, na.rm = TRUE)

## [1] 4.602416
mean(df_end$matchmaker_3, na.rm = TRUE)

## [1] 0.708502
mean(df_end$bl_mother_schooled, na.rm = TRUE)

## [1] 0.6350053
mean(df_end$bl_education, na.rm = TRUE)

## [1] 7.052062
mean(df_end$older_sister, na.rm = TRUE)

## [1] 0.1224602
mean(df_end$bl_desired_marriage_age, na.rm = TRUE)

## [1] 19.42449
# older_sister par groupe de traitement
aggregate(older_sister ~ treatment_type, data = df_end,
          FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1))

##   treatment_type older_sister
## 1                1          12.5
## 2                2          13.2
## 3                3          11.0
## 4                4          12.2
# bl_age_reported par groupe de traitement
aggregate(bl_age_reported ~ treatment_type, data = df_end,
          FUN = function(x) round(mean(x, na.rm=TRUE), 2))

##   treatment_type bl_age_reported
## 1                1          15.72
## 2                2          15.67
## 3                3          15.69
## 4                4          15.73
# bl_still_in_school par groupe de traitement
aggregate(bl_still_in_school ~ treatment_type, data = df_end,
          FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1))

##   treatment_type bl_still_in_school
## 1                1          63.3
## 2                2          65.0
## 3                3          66.0
## 4                4          62.3
# bl_HHsize par groupe de traitement
aggregate(bl_HHsize ~ treatment_type, data = df_end,
          FUN = function(x) round(mean(x, na.rm=TRUE), 2))

```

```
## treatment_type bl_HHsize
## 1 1 4.45
## 2 2 4.70
## 3 3 4.78
## 4 4 4.60

# matchmaker_3 par groupe de traitement
aggregate(matchmaker_3 ~ treatment_type, data = df_end,
           FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1))
```

```
## treatment_type matchmaker_3
## 1 1 73.6
## 2 2 60.5
## 3 3 70.4
## 4 4 74.2

# bl_mother_schooled par groupe de traitement
aggregate(bl_mother_schooled ~ treatment_type, data = df_end,
           FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1))
```

```
## treatment_type bl_mother_schooled
## 1 1 62.1
## 2 2 66.5
## 3 3 65.4
## 4 4 62.0

# bl_education par groupe de traitement
aggregate(bl_education ~ treatment_type, data = df_end,
           FUN = function(x) round(mean(x, na.rm=TRUE), 2))
```

```
## treatment_type bl_education
## 1 1 7.09
## 2 2 7.19
## 3 3 7.07
## 4 4 6.94

# bl_desired_marriage_age par groupe de traitement
aggregate(bl_desired_marriage_age ~ treatment_type, data = df_end,
           FUN = function(x) round(mean(x, na.rm=TRUE), 2))
```

```
## treatment_type bl_desired_marriage_age
## 1 1 19.78
## 2 2 19.42
## 3 3 19.19
## 4 4 19.22
```

Test des égalités de moyenne dans le tableau de balance

```
# Liste de tes variables
vars_cont <- c("bl_age_reported", "bl_education", "bl_HHsize", "bl_desired_marriage_age")
vars_cat <- c("bl_still_in_school", "bl_mother_schooled", "older_sister")
vars_all <- c(vars_cont, vars_cat)

# P-value via régression OLS avec erreurs-types clusterisées
p_clustered <- sapply(vars_all, function(v) {
  formula <- as.formula(paste(v, "~ factor(treatment_type)"))
  reg <- lm(formula, data = df_end)
```

```

# Test F avec erreurs-types clusterisées au niveau CLUSTER
ct <- coeftest(reg, vcov = vcovCL(reg, cluster = ~CLUSTER))
# P-value du test joint (F-test)
wt <- waldtest(reg, vcov = vcovCL(reg, cluster = ~CLUSTER))
round(wt$`Pr(>F)`[2], 3)
})

print(p_clustered)

##          bl_age_reported          bl_education          bl_HHsize
##              0.629              0.750              0.448
## bl_desired_marriage_age    bl_still_in_school    bl_mother_schooled
##              0.348              0.732              0.771
##          older_sister
##              0.848

```

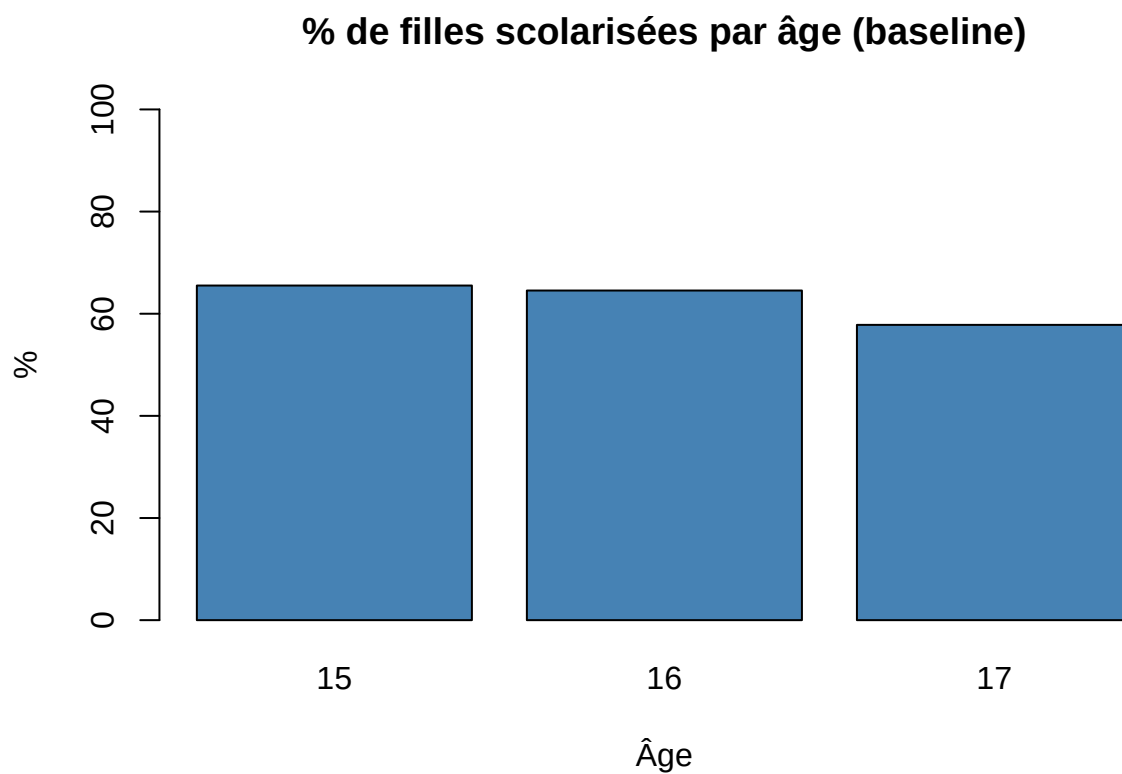
Visualisations supplémentaires pour la baseline

```

# Créer tranche_age dans df_end
df_end$tranche_age <- cut(df_end$bl_age_reported,
                          breaks = c(14, 15, 16, 17),
                          labels = c("15", "16", "17"),
                          include.lowest = TRUE)

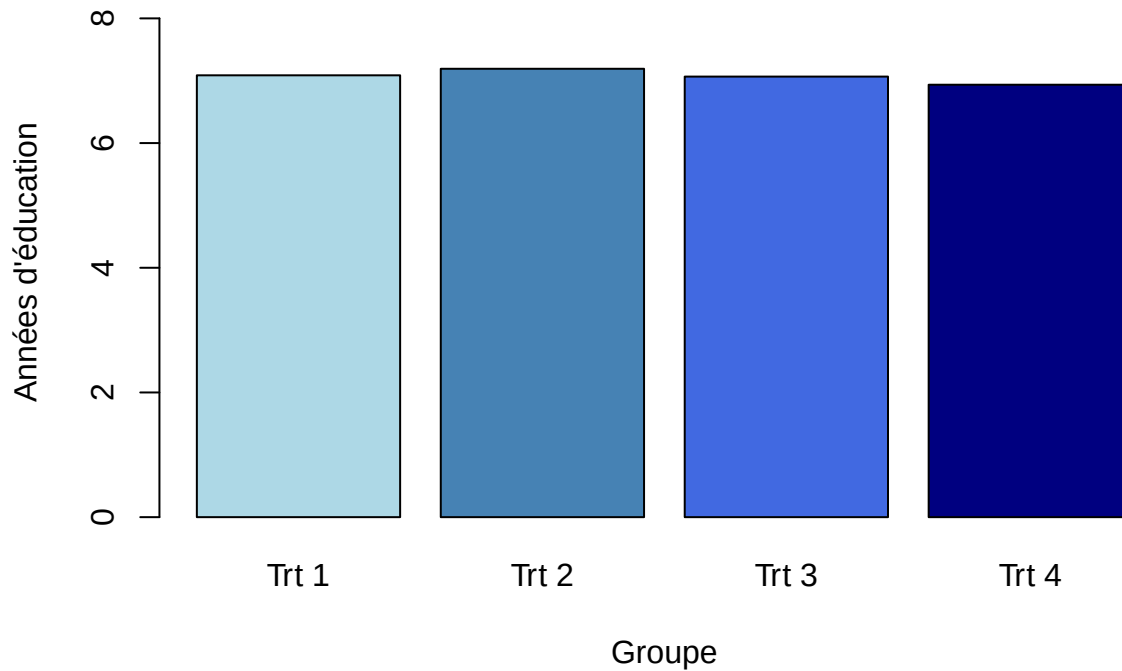
scol <- tapply(df_end$bl_still_in_school, df_end$tranche_age,
               mean, na.rm = TRUE) * 100
barplot(scol,
        main = "% de filles scolarisées par âge (baseline)",
        xlab = "Âge", ylab = "%",
        col = "steelblue", ylim = c(0, 100))

```

```
educ_trt <- aggregate(bl_education ~ treatment_type,  
                      data = df_end, FUN = mean)  
barplot(educ_trt$bl_education,  
        names.arg = paste("Trt", educ_trt$treatment_type),  
        main = "Éducation moyenne par groupe de traitement (baseline)",  
        xlab = "Groupe", ylab = "Années d'éducation",  
        col = c("lightblue", "steelblue", "royalblue", "navy"),  
        ylim = c(0, max(educ_trt$bl_education) + 1))
```

Éducation moyenne par groupe de traitement (baseline)



Endline : Stats par groupe de traitement

```
# Outcomes principaux par groupe
aggregate(ever_married ~ treatment_type, data = df_end,
          FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1))
```

```
##   treatment_type ever_married
## 1             1         90.8
## 2             2         90.6
## 3             3         87.8
## 4             4         89.8
```

```
aggregate(still_in_school ~ treatment_type, data = df_end,
          FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1))
```

```
##   treatment_type still_in_school
## 1             1          18.0
## 2             2          12.9
## 3             3          19.8
## 4             4          17.8
```

```
aggregate(age_married_desired ~ treatment_type, data = df_end,
          FUN = mean, na.rm = TRUE)
```

```
##   treatment_type age_married_desired
## 1             1         20.37485
## 2             2         20.16946
```

```
## 3          3          20.36102
## 4          4          20.23231
```

```
aggregate(secondary_complete ~ treatment_type, data = df_end,
          FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1))
```

```
## treatment_type secondary_complete
## 1          1          38.4
## 2          2          32.6
## 3          3          34.8
## 4          4          34.9
```

Commentaire - Outcome principaux par groupe : A la fin de l'expérience, on remarque que le groupe ayant reçu le troisième traitement présente un taux plus faible de filles mariées. Le troisième traitement correspond à l'incitation financière combinée à l'empowerment (1 : Empowerment, 2 : Incentive ; Traitement 4 : groupe de contrôle)

```
# bl_HHsize : ménage de la fille au début
cat("Taille ménage moyenne :", round(mean(df_end$bl_HHsize, na.rm=TRUE), 2), "\n")
```

```
## Taille ménage moyenne : 4.6
```

```
cat("(Variable mesurée à la baseline = ménage d'origine de la fille)\n")
```

```
## (Variable mesurée à la baseline = ménage d'origine de la fille)
```

```
# older_sister par groupe de traitement
aggregate(older_sister ~ treatment_type, data = df_end,
          FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1))
```

```
## treatment_type older_sister
## 1          1          12.5
## 2          2          13.2
## 3          3          11.0
## 4          4          12.2
```

```
# matchmaker_3, has_primary, has_secondary, bl_mother_schooled
vars_infra <- c("matchmaker_3", "has_primary", "has_secondary", "bl_mother_schooled")
sapply(df_end[, vars_infra], function(x) round(mean(x, na.rm=TRUE)*100, 1))
```

```
## matchmaker_3      has_primary      has_secondary bl_mother_schooled
##          70.9          52.0          26.8          63.5
```

Commentaire : 70.9% des ménages possèdent trois matchmaker ou plus. A l'endline, moitié des filles ont complété l'éducation primaire et un peu plus d'un quart le secondaire.

```
# Évolution de la scolarisation baseline / endline par groupe de traitement
```

```
# % scolarisées baseline par traitement
scol_bl <- aggregate(bl_still_in_school ~ treatment_type,
                    data = df_end,
                    FUN = function(x) mean(x, na.rm = TRUE) * 100)$bl_still_in_school
```

```
# % scolarisées endline par traitement
scol_el <- aggregate(still_in_school ~ treatment_type,
                    data = df_end,
                    FUN = function(x) mean(x, na.rm = TRUE) * 100)$still_in_school
```

```
df_plot_scol <- data.frame(
```

```

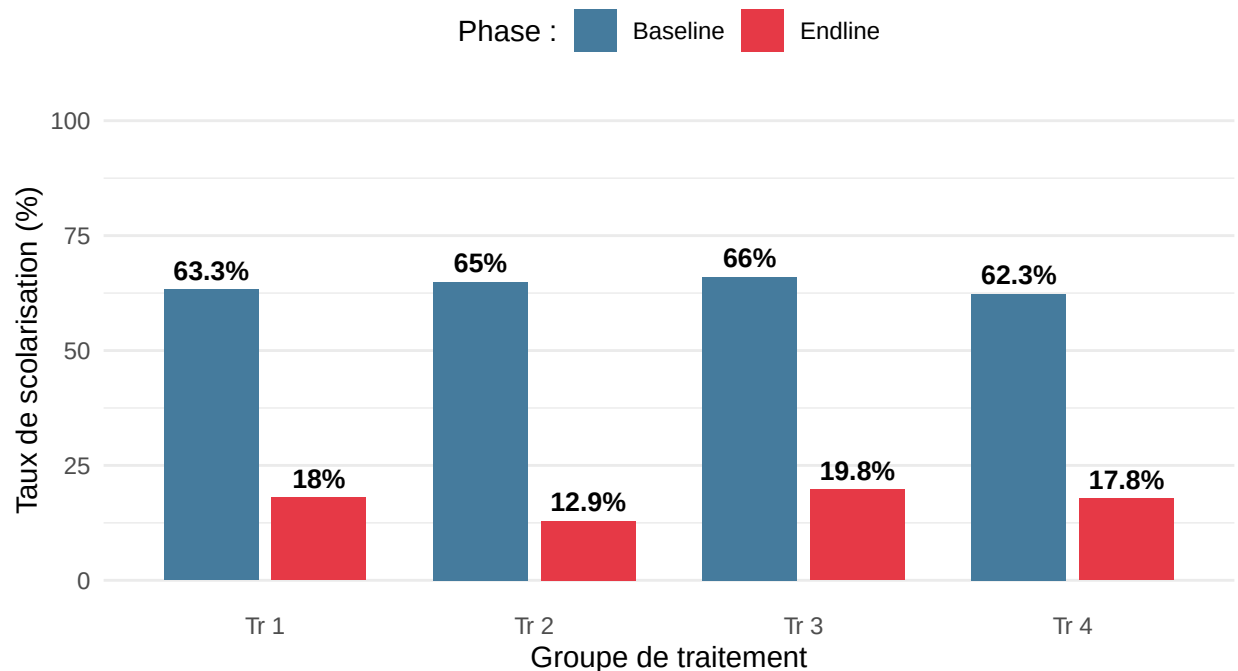
Traitement = paste("Tr", 1:4),
Baseline = scol_bl,
Endline = scol_el
) %>%
pivot_longer(cols = c(Baseline, Endline), names_to = "Temps", values_to = "Pourcentage")

ggplot(df_plot_scol, aes(x = Traitement, y = Pourcentage, fill = Temps)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = paste0(round(Pourcentage, 1), "%"),
    position = position_dodge(width = 0.8), vjust = -0.5, size = 3.5, fontface = "bold") +
  scale_fill_manual(values = c("Baseline" = "#457b9d", "Endline" = "#e63946")) +
  labs(title = "Évolution de la scolarisation par groupe de traitement",
    subtitle = "Comparaison entre le début (baseline) et la fin (endline) de l'étude",
    x = "Groupe de traitement", y = "Taux de scolarisation (%)",
    caption = "Échantillon : 1821 filles âgées de 15 à 17 ans non mariées au début de l'étude.\nTr 1
    fill = "Phase :") +
  theme_minimal() +
  theme(legend.position = "top",
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.major.x = element_blank()) +
  ylim(0, 100)

```

Évolution de la scolarisation par groupe de traitement

Comparaison entre le début (baseline) et la fin (endline) de l'étude



Échantillon : 1821 filles âgées de 15 à 17 ans non mariées au début de l'étude.
 Tr 1 : Empowerment | Tr 2 : Incentive | Tr 3 : Empowerment + Incentive | Tr 4 : Contrôle

```
# Taux de mariage par âge baseline (endline)
```

```
prop_mar <- tapply(df_end$ever_married,
```

```

        df_end$bl_age_reported,
        mean,
        na.rm = TRUE) * 100

df_mar_age <- data.frame(
  age_baseline = names(prop_mar),
  pourcentage = as.numeric(prop_mar)
)

ggplot(df_mar_age, aes(x = age_baseline, y = pourcentage)) +
  geom_bar(stat = "identity", width = 0.6, fill = "steelblue") +

  geom_text(aes(label = paste0(round(pourcentage, 1), "%")),
            vjust = -0.8, size = 4, fontface = "bold") +

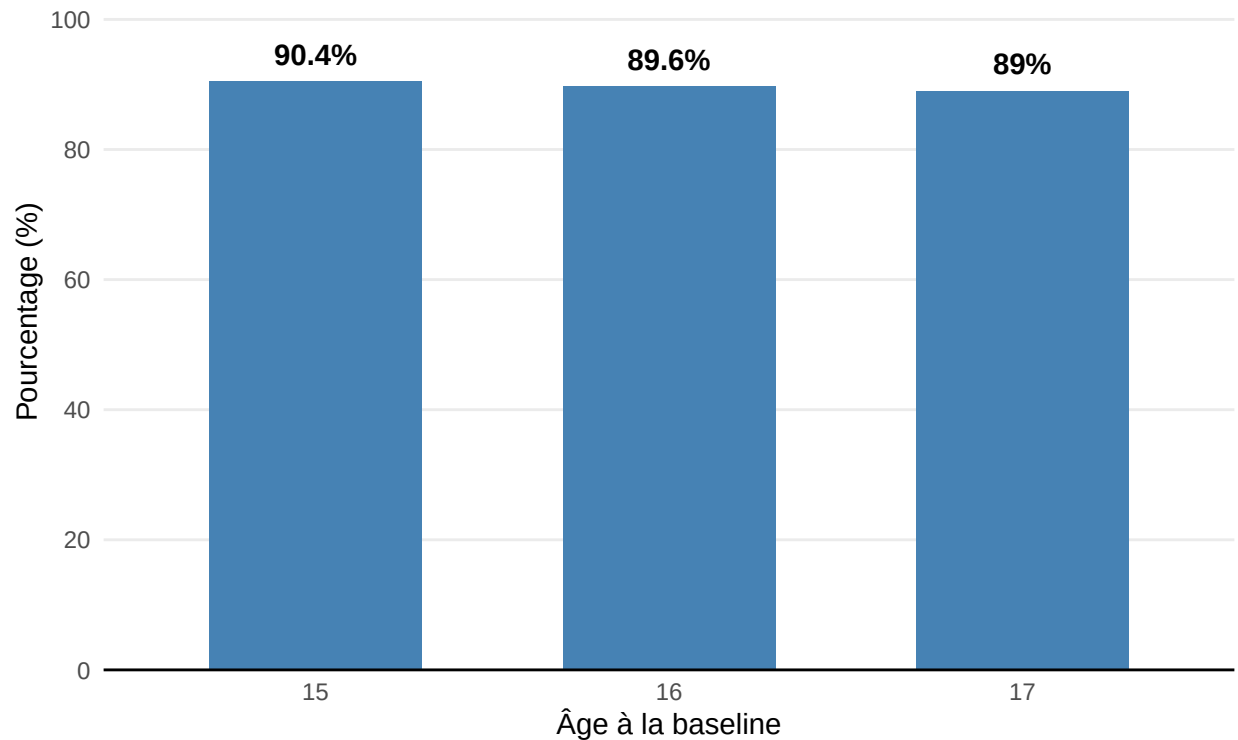
  scale_y_continuous(limits = c(0, 105),
                     breaks = seq(0, 100, 20),
                     expand = expansion(mult = c(0, 0.05))) +

  theme_minimal() +
  labs(title = "Taux de mariage à l'endline par âge initial",
       x = "Âge à la baseline",
       y = "Pourcentage (%)") +

  theme(
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    axis.line.x = element_line(color = "black") # Souligne l'axe X pour plus de propreté
  )

```

Taux de mariage à l'endline par âge initial



Statistiques à la endline

`names(df_2)`

```
## [1] "girlID" "HHID"
## [3] "CLUSTER" "bl_age_reported"
## [5] "unionID" "washedout"
## [7] "treatment_type" "at1"
## [9] "at2" "at3"
## [11] "at4" "third"
## [13] "bl_ever_married" "ss_girlID"
## [15] "access_cash" "allow_dress"
## [17] "angry_cellphone" "angry_for_food"
## [19] "angry_for_friend" "char_marriage_age"
## [21] "char_marriage_character" "char_marriage_edu"
## [23] "char_marriage_family" "char_marriage_looks"
## [25] "char_marriage_nature" "char_marriage_other"
## [27] "char_marriage_potential" "char_marriage_religious"
## [29] "char_marriage_reputation" "char_marriage_responsible"
## [31] "char_marriage_romantic" "char_marriage_wealth"
## [33] "education_resources" "hus_income"
## [35] "husband_level" "iga_current"
## [37] "latest_marry" "menstruationunable_act"
## [39] "met_hus_pre_mar" "endline"
## [41] "bl_still_in_school" "dowry"
## [43] "c_marriage_age" "arranged_marriage"
```

## [45]	"bl_public_transit"	"anyoil"
## [47]	"anyemp"	"oil_kk"
## [49]	"oil"	"oil_sheet"
## [51]	"anylist"	"list_kk"
## [53]	"ever_married"	"marriage_age1"
## [55]	"marriage_age3"	"marriage_age"
## [57]	"before_miss"	"under_18"
## [59]	"under_16"	"still_in_school"
## [61]	"education"	"ever_birth_20"
## [63]	"majorpurchase_dec"	"accessories_dec"
## [65]	"bed_dec"	"bedsheet_dec"
## [67]	"cosmetics_dec"	"phone_recharge_dec"
## [69]	"utensils_dec"	"budget_dec"
## [71]	"loan_dec"	"ornament_dec"
## [73]	"ever_worked"	"denmeher"
## [75]	"pregnancy_risks"	"husband_age"
## [77]	"husband_edu"	"hus_secondary_complete"
## [79]	"hus_formal"	"bl_education_mother"
## [81]	"bl_HHsize"	"older_sister"
## [83]	"bl_age6"	"bl_age7"
## [85]	"bl_age8"	"bl_age9"
## [87]	"bl_age10"	"bl_age11"
## [89]	"bl_age12"	"bl_age13"
## [91]	"bl_age14"	"bl_age15"
## [93]	"bl_age16"	"bl_age17"
## [95]	"bl_mother_schooled"	"distance_vc_ss"
## [97]	"distance_hh_ss"	"vill_radius_500_oil"
## [99]	"distance_hh_oil"	"village_distance"
## [101]	"distance_hh_vc"	"distance_hh_nvc"
## [103]	"kk_attend"	"kk_member"
## [105]	"oil_heard"	"oil_received"
## [107]	"ml_ever_married"	"received_oil_il"
## [109]	"bl_highest_education"	"bl_allow_study"
## [111]	"bl_allow_dress"	"bl_allow_makeup"
## [113]	"bl_husband_level"	"bl_educ_goal_comfort"
## [115]	"bl_puberty_comfort"	"bl_marriage_time_comfort"
## [117]	"bl_marriage_choice_comfort"	"bl_marriage_dowry_comfort"
## [119]	"bl_harrassment_discuss"	"bl_harrassment_comfort"
## [121]	"bl_earliest_marry"	"bl_earliest_marry_why_phys"
## [123]	"bl_latest_marry"	"bl_desired_marriage_age"
## [125]	"bl_dowry_amnt"	"bl_reason_occupation_none"
## [127]	"bl_reason_occupation_emerg"	"bl_stunted"
## [129]	"bl_bmi"	"would_tell"
## [131]	"no_menstru"	"wives_less_educated"
## [133]	"girls_study_far"	"education_desired"
## [135]	"age_married_desired"	"max_marriage_age"
## [137]	"bl_income"	"high_p_cons"
## [139]	"girls_washedout"	"older_sister_miss"
## [141]	"bl_still_in_school_miss"	"bl_education_mother_miss"
## [143]	"bl_HHsize_miss"	"bl_public_transit_miss"
## [145]	"bl_bmi_miss"	"bl_stunted_miss"
## [147]	"bl_income_miss"	"g_cons_index"
## [149]	"high_g_cons"	"distance_hh_oil_miss"
## [151]	"g_cons_index_miss"	

```

mean(df_end$age_gap, na.rm = TRUE)

## [1] 5.379111
mean(df_end$outside_village, na.rm = TRUE)

## [1] 0.906087
mean(df_end$under_18, na.rm = TRUE)

## [1] 0.5336506
mean(df_end$arranged_marriage, na.rm = TRUE)

## [1] 0.9088542
mean(df_end$ever_married, na.rm = TRUE)

## [1] 0.8988391
mean(df_end$ever_birth_20, na.rm = TRUE)

## [1] 0.2767956
mean(df_end$marriage_age, na.rm = TRUE)

## [1] 17.3227
mean(df_end$el_age_predicted, na.rm = TRUE)    #23.88

## Warning in mean.default(df_end$el_age_predicted, na.rm = TRUE): argument is not
## numeric or logical: returning NA

## [1] NA

# Variables binaires : pourcentage
vars_pct <- c("outside_village", "under_18", "arranged_marriage",
              "ever_married", "ever_birth_20", "ever_worked")

for (v in vars_pct) {
  cat(v, ":", round(mean(df_end[[v]], na.rm = TRUE) * 100, 1), "%\n")
}

## outside_village : 90.6 %
## under_18 : 53.4 %
## arranged_marriage : 90.9 %
## ever_married : 89.9 %
## ever_birth_20 : 27.7 %
## ever_worked : 56 %

# Variables continues : moyenne (sd)
vars_cont <- c("age_gap", "marriage_age", "el_age_predicted", "bl_income")

for (v in vars_cont) {
  cat(v, ":", round(mean(df_end[[v]], na.rm = TRUE), 2),
      "(", round(sd(df_end[[v]], na.rm = TRUE), 2), ")\n")
}

## age_gap : 5.38 ( 4.35 )
## marriage_age : 17.32 ( 2.49 )

## Warning in mean.default(df_end[[v]], na.rm = TRUE): argument is not numeric or

```



```
## logical: returning NA
## el_age_predicted : NA ( NA )
## bl_income : 19.63 ( 19.65 )
```

Selon l'indice de conservatisme

```
# Variables d'outcome à analyser
vars_outcome <- c("ever_married", "under_18", "still_in_school",
                  "secondary_complete", "age_married_desired",
                  "ever_birth_20", "ever_worked")

# ***- 1. CONSERVATISME DE LA FILLE (high_g_cons) ***-

cat("=== CONSERVATISME DE LA FILLE (high_g_cons) ===\n\n")

## === CONSERVATISME DE LA FILLE (high_g_cons) ===

# Variables binaires + proportions
for (v in vars_outcome) {
  cat(v, ":\n")
  print(aggregate(df_end[[v]] ~ df_end$high_g_cons,
                  FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1)))
  cat("\n")
}
```

```
## ever_married :
##   df_end$high_g_cons df_end[[v]]
## 1                   0         86.9
## 2                   1         94.1
##
## under_18 :
##   df_end$high_g_cons df_end[[v]]
## 1                   0         46.0
## 2                   1         63.1
##
## still_in_school :
##   df_end$high_g_cons df_end[[v]]
## 1                   0         25.0
## 2                   1          6.9
##
## secondary_complete :
##   df_end$high_g_cons df_end[[v]]
## 1                   0         50.7
## 2                   1         15.7
##
## age_married_desired :
##   df_end$high_g_cons df_end[[v]]
## 1                   0        2095.2
## 2                   1        1936.6
##
## ever_birth_20 :
##   df_end$high_g_cons df_end[[v]]
## 1                   0         23.2
## 2                   1         33.9
```

```

##
## ever_worked :
##   df_end$high_g_cons df_end[[v]]
## 1           0       58.2
## 2           1       53.0

# Variable continue : bl_income
cat("bl_income :\n")

## bl_income :
aggregate(bl_income ~ high_g_cons, data = df_end,
          FUN = function(x) c(mean = round(mean(x, na.rm=TRUE), 2),
                               sd   = round(sd(x, na.rm=TRUE), 2)))

##   high_g_cons bl_income.mean bl_income.sd
## 1           0       21.15      23.39
## 2           1       17.53      12.51

# ***- 2. CONSERVATISME DES PARENTS (high_p_cons) ***-

cat("\n=== CONSERVATISME DES PARENTS (high_p_cons) ===\n\n")

##
## === CONSERVATISME DES PARENTS (high_p_cons) ===

# Variables binaires + proportions
for (v in vars_outcome) {
  cat(v, ":\n")
  print(aggregate(df_end[[v]] ~ df_end$high_p_cons,
                  FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1)))
  cat("\n")
}

## ever_married :
##   df_end$high_p_cons df_end[[v]]
## 1           0       87.1
## 2           1       93.9
##
## under_18 :
##   df_end$high_p_cons df_end[[v]]
## 1           0       45.5
## 2           1       63.4
##
## still_in_school :
##   df_end$high_p_cons df_end[[v]]
## 1           0       23.8
## 2           1        8.8
##
## secondary_complete :
##   df_end$high_p_cons df_end[[v]]
## 1           0       49.2
## 2           1       18.8
##
## age_married_desired :
##   df_end$high_p_cons df_end[[v]]
## 1           0      2147.2

```

```
## 2          1      1874.8
##
## ever_birth_20 :
##   df_end$high_p_cons df_end[[v]]
## 1          0       21.7
## 2          1       35.6
##
## ever_worked :
##   df_end$high_p_cons df_end[[v]]
## 1          0       57.5
## 2          1       54.3

# Variable continue : bl_income
cat("bl_income : \n")

## bl_income :

aggregate(bl_income ~ high_p_cons, data = df_end,
          FUN = function(x) c(mean = round(mean(x, na.rm=TRUE), 2),
                               sd = round(sd(x, na.rm=TRUE), 2)))

##   high_p_cons bl_income.mean bl_income.sd
## 1          0       21.62      23.63
## 2          1       17.82      12.30

# ***- 3. Tableau récapitulatif ***-

cat("\n=== TABLEAU RÉCAP ===\n")

##
## === TABLEAU RÉCAP ===

recap_cons <- data.frame(
  Variable = vars_outcome,
  g_cons_0 = sapply(vars_outcome, function(v)
    round(mean(df_end[[v]][df_end$high_g_cons == 0], na.rm=TRUE)*100, 1)),
  g_cons_1 = sapply(vars_outcome, function(v)
    round(mean(df_end[[v]][df_end$high_g_cons == 1], na.rm=TRUE)*100, 1)),
  p_cons_0 = sapply(vars_outcome, function(v)
    round(mean(df_end[[v]][df_end$high_p_cons == 0], na.rm=TRUE)*100, 1)),
  p_cons_1 = sapply(vars_outcome, function(v)
    round(mean(df_end[[v]][df_end$high_p_cons == 1], na.rm=TRUE)*100, 1))
)
colnames(recap_cons) <- c("Variable",
  "Fille peu conservatrice",
  "Fille très conservatrice",
  "Parents peu conservateurs",
  "Parents très conservateurs")
print(recap_cons)

##
##          Variable Fille peu conservatrice
## ever_married      ever_married      86.9
## under_18          under_18      46.0
## still_in_school    still_in_school    25.0
## secondary_complete secondary_complete    50.7
## age_married_desired age_married_desired 2095.2
## ever_birth_20      ever_birth_20     23.2
```

```
## ever_worked          ever_worked          58.2
##          Fille très conservatrice Parents peu conservateurs
## ever_married          94.1          87.1
## under_18          63.1          45.5
## still_in_school          6.9          23.8
## secondary_complete          15.7          49.2
## age_married_desired          1936.6          2147.2
## ever_birth_20          33.9          21.7
## ever_worked          53.0          57.5
##          Parents très conservateurs
## ever_married          93.9
## under_18          63.4
## still_in_school          8.8
## secondary_complete          18.8
## age_married_desired          1874.8
## ever_birth_20          35.6
## ever_worked          54.3
```

```
# bl_income séparément (variable continue)
cat("\nbl_income selon conservatisme :\n")
```

```
##
```

```
## bl_income selon conservatisme :
```

```
data.frame(
  Groupe = c("Fille peu cons.", "Fille très cons.",
             "Parents peu cons.", "Parents très cons."),
  Moyenne = c(
    round(mean(df_end$bl_income[df_end$high_g_cons == 0], na.rm=TRUE), 2),
    round(mean(df_end$bl_income[df_end$high_g_cons == 1], na.rm=TRUE), 2),
    round(mean(df_end$bl_income[df_end$high_p_cons == 0], na.rm=TRUE), 2),
    round(mean(df_end$bl_income[df_end$high_p_cons == 1], na.rm=TRUE), 2)
  ),
  SD = c(
    round(sd(df_end$bl_income[df_end$high_g_cons == 0], na.rm=TRUE), 2),
    round(sd(df_end$bl_income[df_end$high_g_cons == 1], na.rm=TRUE), 2),
    round(sd(df_end$bl_income[df_end$high_p_cons == 0], na.rm=TRUE), 2),
    round(sd(df_end$bl_income[df_end$high_p_cons == 1], na.rm=TRUE), 2)
  )
)
```

```
##          Groupe Moyenne    SD
## 1  Fille peu cons.   21.15 23.39
## 2  Fille très cons.   17.53 12.51
## 3  Parents peu cons.   21.62 23.63
## 4  Parents très cons.   17.82 12.30
```

Statistiques à la endline pour le groupe de contrôle seulement

```
# =====
# RESTRICTION AU GROUPE CONTRÔLE (treatment_type == 4)
# =====

df_control <- subset(df_end, treatment_type == 4)
```

```

cat("Taille de l'échantillon Contrôle :", nrow(df_control), "\n\n")

## Taille de l'échantillon Contrôle : 597
# *** Stats descriptives de base ***
vars_pct <- c("outside_village", "under_18", "arranged_marriage",
             "ever_married", "ever_birth_20", "ever_worked")

cat("=== POURCENTAGES (CONTRÔLE) ===\n")

## === POURCENTAGES (CONTRÔLE) ===
for (v in vars_pct) {
  cat(v, ":", round(mean(df_control[[v]], na.rm = TRUE) * 100, 1), "%\n")
}

## outside_village : 90.5 %
## under_18 : 52.2 %
## arranged_marriage : 90.8 %
## ever_married : 89.8 %
## ever_birth_20 : 27.5 %
## ever_worked : 54.3 %

vars_cont <- c("age_gap", "marriage_age", "el_age_predicted", "bl_income")

cat("\n=== MOYENNES ET SD (CONTRÔLE) ===\n")

##
## === MOYENNES ET SD (CONTRÔLE) ===
for (v in vars_cont) {
  cat(v, ":", round(mean(df_control[[v]], na.rm = TRUE), 2),
      "(", round(sd(df_control[[v]], na.rm = TRUE), 2), ")\n")
}

## age_gap : 5.48 ( 4.39 )
## marriage_age : 17.34 ( 2.55 )

## Warning in mean.default(df_control[[v]], na.rm = TRUE): argument is not numeric
## or logical: returning NA

## el_age_predicted : NA ( NA )
## bl_income : 20.27 ( 25.9 )

# =====
# STATS SELON L'INDICE DE CONSERVATISME (DANS LE CONTRÔLE)
# =====

vars_outcome <- c("ever_married", "under_18", "still_in_school",
                 "secondary_complete", "age_married_desired",
                 "ever_birth_20", "ever_worked")

# ***- 1. CONSERVATISME DE LA FILLE ***-
cat("\n=== CONSERVATISME DE LA FILLE (high_g_cons) - GROUPE CONTRÔLE ===\n")

##
## === CONSERVATISME DE LA FILLE (high_g_cons) - GROUPE CONTRÔLE ===

```

```

for (v in vars_outcome) {
  cat(v, ":\n")
  print(aggregate(df_control[[v]] ~ df_control$high_g_cons,
    FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1)))
}

```

```

## ever_married :
##   df_control$high_g_cons df_control[[v]]
## 1                      0             85.2
## 2                      1             95.8
## under_18 :
##   df_control$high_g_cons df_control[[v]]
## 1                      0             41.4
## 2                      1             65.7
## still_in_school :
##   df_control$high_g_cons df_control[[v]]
## 1                      0             26.9
## 2                      1              6.2
## secondary_complete :
##   df_control$high_g_cons df_control[[v]]
## 1                      0             51.4
## 2                      1             14.4
## age_married_desired :
##   df_control$high_g_cons df_control[[v]]
## 1                      0            2088.1
## 2                      1            1939.7
## ever_birth_20 :
##   df_control$high_g_cons df_control[[v]]
## 1                      0             24.9
## 2                      1             30.9
## ever_worked :
##   df_control$high_g_cons df_control[[v]]
## 1                      0             57.9
## 2                      1             49.8

```

*# ***- 2. CONSERVATISME DES PARENTS ***-*

```
cat("\n=== CONSERVATISME DES PARENTS (high_p_cons) - GROUPE CONTRÔLE ===\n")
```

```
##
```

```
## === CONSERVATISME DES PARENTS (high_p_cons) - GROUPE CONTRÔLE ===
```

```

for (v in vars_outcome) {
  cat(v, ":\n")
  print(aggregate(df_control[[v]] ~ df_control$high_p_cons,
    FUN = function(x) round(mean(x, na.rm=TRUE)*100, 1)))
}

```

```

## ever_married :
##   df_control$high_p_cons df_control[[v]]
## 1                      0             87.0
## 2                      1             93.9
## under_18 :
##   df_control$high_p_cons df_control[[v]]
## 1                      0             43.9
## 2                      1             63.0

```

```
## still_in_school :
##   df_control$high_p_cons df_control[[v]]
## 1                0        24.1
## 2                1         9.0
## secondary_complete :
##   df_control$high_p_cons df_control[[v]]
## 1                0        48.2
## 2                1       17.5
## age_married_desired :
##   df_control$high_p_cons df_control[[v]]
## 1                0       2127.6
## 2                1       1882.2
## ever_birth_20 :
##   df_control$high_p_cons df_control[[v]]
## 1                0        22.9
## 2                1        34.0
## ever_worked :
##   df_control$high_p_cons df_control[[v]]
## 1                0        52.6
## 2                1        56.7

# ***- 3. TABLEAU RÉCAP (CONTRÔLE UNIQUEMENT) ***-
cat("\n=== TABLEAU RÉCAPITULATIF (GROUPE CONTRÔLE) ===\n")

##
## === TABLEAU RÉCAPITULATIF (GROUPE CONTRÔLE) ===

recap_cons_control <- data.frame(
  Variable = vars_outcome,
  g_cons_0 = apply(vars_outcome, function(v)
    round(mean(df_control[[v]][df_control$high_g_cons == 0], na.rm=TRUE)*100, 1)),
  g_cons_1 = apply(vars_outcome, function(v)
    round(mean(df_control[[v]][df_control$high_g_cons == 1], na.rm=TRUE)*100, 1)),
  p_cons_0 = apply(vars_outcome, function(v)
    round(mean(df_control[[v]][df_control$high_p_cons == 0], na.rm=TRUE)*100, 1)),
  p_cons_1 = apply(vars_outcome, function(v)
    round(mean(df_control[[v]][df_control$high_p_cons == 1], na.rm=TRUE)*100, 1))
)

colnames(recap_cons_control) <- c("Variable",
  "Fille peu cons. (C)",
  "Fille très cons. (C)",
  "Parents peu cons. (C)",
  "Parents très cons. (C)")

print(recap_cons_control)

##
##           Variable Fille peu cons. (C)
## ever_married      ever_married      85.2
## under_18          under_18      41.4
## still_in_school   still_in_school   26.9
## secondary_complete secondary_complete  51.4
## age_married_desired age_married_desired 2088.1
## ever_birth_20      ever_birth_20    24.9
## ever_worked        ever_worked     57.9
##           Fille très cons. (C) Parents peu cons. (C)
```

```
## ever_married          95.8          87.0
## under_18              65.7          43.9
## still_in_school       6.2          24.1
## secondary_complete    14.4          48.2
## age_married_desired   1939.7        2127.6
## ever_birth_20         30.9          22.9
## ever_worked           49.8          52.6
##                      Parents très cons. (C)
## ever_married          93.9
## under_18              63.0
## still_in_school       9.0
## secondary_complete    17.5
## age_married_desired   1882.2
## ever_birth_20         34.0
## ever_worked           56.7
```

```
# bl_income séparément
cat("\nbl_income selon conservatisme (CONTRÔLE) :\n")
```

```
##
## bl_income selon conservatisme (CONTRÔLE) :
print(data.frame(
  Groupe = c("Fille peu cons.", "Fille très cons.", "Parents peu cons.", "Parents très cons."),
  Moyenne = c(
    round(mean(df_control$bl_income[df_control$high_g_cons == 0], na.rm=TRUE), 2),
    round(mean(df_control$bl_income[df_control$high_g_cons == 1], na.rm=TRUE), 2),
    round(mean(df_control$bl_income[df_control$high_p_cons == 0], na.rm=TRUE), 2),
    round(mean(df_control$bl_income[df_control$high_p_cons == 1], na.rm=TRUE), 2)
  ),
  SD = c(
    round(sd(df_control$bl_income[df_control$high_g_cons == 0], na.rm=TRUE), 2),
    round(sd(df_control$bl_income[df_control$high_g_cons == 1], na.rm=TRUE), 2),
    round(sd(df_control$bl_income[df_control$high_p_cons == 0], na.rm=TRUE), 2),
    round(sd(df_control$bl_income[df_control$high_p_cons == 1], na.rm=TRUE), 2)
  )
))
```

```
##           Groupe Moyenne    SD
## 1  Fille peu cons.   21.88 32.31
## 2  Fille très cons.   18.21 13.72
## 3  Parents peu cons.  21.97 31.84
## 4  Parents très cons. 17.91 13.67
```